

JEE MAIN REVISION MASTER: SOLID STATE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Crystal Lattices, Unit Cells & Density Formula	14 PYQs	Crystallography
High	Point Defects & Magnetic Properties of Solids	9 PYQs	Crystal Imperfections

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Crystal Lattices, Unit Cells & Density Formula	$d = \frac{Z \cdot M}{a^3 \cdot N_A}, \quad \text{BCC: } 4r = \sqrt{3}a, \quad \text{FCC: } 4r = \sqrt{2}a, \quad Z_{\text{fcc}} = 4, \quad Z_{\text{bcc}} = 2$
Point Defects & Magnetic Properties of Solids	$\text{Schottky: Density Decreases}, \quad \text{Frenkel: Density Unchanged}, \quad \text{AgBr: Shows Both Defects}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Crystal Lattices, Unit Cells & Density Formula (High Yield — 14 PYQs)

Core Principle: Unit cells: Simple cubic ($Z=1$, coordination number 6, packing efficiency 52.4%, $r = a/2$). BCC ($Z=2$, coordination number 8, packing efficiency 68%, $r = \frac{\sqrt{3}}{4}a$). FCC ($Z=4$, coordination number 12, packing efficiency 74%, $r = \frac{a\sqrt{2}}{4}$). Density formula $d = (Z \cdot M / a^3 \cdot N_A)$.

■ **Caution / JEE Pitfall:** Edge length a in the density formula must be converted to cm ($1 \text{ pm} = 10^{-10} \text{ cm}$) to obtain density in g/cm^3 !

■ **Speed Trick:** Packing efficiency order: FCC (74%) > BCC (68%) > SC (52.4%). Space unoccupied (voids) in FCC is 26%, in BCC is 32%, in SC is 47.6%.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following crystals has the unit cell such that $a=b \neq c$ and $\alpha=\beta=90^\circ$, $\gamma=120^\circ$?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "A metal crystallises in a body centred cubic lattice with the metallic radius $\frac{\sqrt{3}}{4}a$. The volume of the unit cell in cm^3 ..." [Ans: N/A]

3.2 Point Defects & Magnetic Properties of Solids (High Yield — 9 PYQs)

Core Principle: Schottky defect: Equal numbers of cations and anions missing implies density decreases (seen in highly ionic salts with high coordination numbers: NaCl, KCl, CsCl). Frenkel defect: Cation displaced to interstitial site implies density remains unchanged (seen in salts with large radius difference: ZnS, AgCl). AgBr exhibits BOTH Schottky and Frenkel defects. F-centers are trapped electrons in anion vacancies that impart color to crystals (e.g. yellow color in NaCl).

■ **Caution / JEE Pitfall:** Frenkel defect does NOT change the density of the crystal; Schottky defect strictly lowers crystal density.

■ **Speed Trick:** AgBr is the classic exam question as the unique compound displaying both Schottky and Frenkel defects.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "In solid state, PCl_5 is a/an..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Match the column A (type of crystalline solid) with the column B (example for each type) A B P. Molecular solid i. SiC Q. Ionic solid ii. Mg..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "Which of the following is not true about the amorphous solids?..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)

- (A) 6.6×10^{20}
- (B) 6.6×10^{23}
- (C) 6.6×10^{19}
- (D) 6.6×10^{22}

Correct Answer: None

Solution Analysis: Given, Face centred crystal, so $Z=4$ edge length $(a)=300 \text{ pm}=300 \times 10^{-10} \text{ cm}$ density $(d)=10 \text{ g cm}^{-3}$
 $N_A=6.022 \times 10^{23}$ Using the formula $d = \frac{Z \times M}{N_A \times a^3}$ $M = \frac{d \times N_A \times a^3}{Z} = \frac{10 \times 6.022 \times 10^{23} \times (300 \times 10^{-10})^3}{4} \Rightarrow M = 40.65 \text{ g}$ So, 40.65 g contains 6.022×10^{23} atoms Let 4.5 g contains x atom $\Rightarrow x = \frac{6.022 \times 10^{23} \times 4.5}{40.65} = 6.6 \times 10^{22}$ atoms

Benchmark Problem 2: KCET 2023 (Q#2)

In solid state, PCl_5 is a/an

- (A) octahedral structure
- (B) ionic solid with $[\text{PCl}_6]^+$ and $[\text{PCl}_4]^-$
- (C) ionic solid with $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$
- (D) covalent solid present in the form of P_2Cl_{10}

Correct Answer: None

Solution Analysis: PCl_5 is an ionic solid composed of $[\text{PCl}_4]^+$ ions and $[\text{PCl}_6]^-$ ions. In an ionic solid, there is a transfer of electrons between atoms, resulting in the formation these cations and anions. These opposite charged ions held together by electrostatic forces forming a crystal lattice structure.